

RATNESH KR. JOSHI

Ph.D. Scholar

"Learning never exhausts the mind."

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EXPERIENCE

Ph.D. Scholar

Indian Institute of Technology Patna

July 2019 – Ongoing Bihta, Bihar, India

- Working on research works focusing on *explainability in dialogue systems* and developing *reinforced conversational agents*.
- Working on analysis of influencing, persuasion, and negotiation along with its ethical applications to develop a conversational agent with such capabilities.
- Worked as a research scholar in a project named *Sevak - An Intelligent Indian Language Chatbot* to develop a dialogue system in both native and Roman scripts for languages Hindi, Bengali, Telugu, English, and Code-mix of all these languages for three domains Railways, Judiciary and Healthcare.
- Worked on GCruz, a sustainability-promoting conversational agent to suggest and influence users to be more environmentally conscious, which involved suggesting and a persuasive system to help support the usability of the suggestion (work done during an internship at Accenture).
- Performed TA duties in three regular courses, viz. Foundations of Machine Learning, Advanced Machine Learning, and Data Structure; and Artificial Intelligence course in Online Executive M.Tech course.

M.Tech. Student.

Central University of Rajasthan, Rajasthan

Aug 2015 – June 2017 Ajmer, Rajasthan, India

- Worked as a researcher on thesis topic of '*Improving Interpretability of Long Short Term Memory (LSTM) with Hidden Markov Models (HMM)*'.
- Performed TA duties in 2 courses, viz. Programming in C for physics and economics by taking labs and exams.

ACADEMIC RESPONSIBILITIES

Teaching Assistant Project Research Scholar
Workshop Organizer Conference Organizer Teaching
Mentoring Interns, Undergraduates and Postgraduates

OTHER ACTIVITIES

Organizer of GIAN Workshop on Deep Learning Techniques for Conversational Artificial Intelligence

Indian Institute of Technology Patna

13 – 24 Apr 2022

Organizer of CEP course on Deep Learning for NLP

Indian Institute of Technology Patna

10 – 23 Jan 2020

ACHIEVEMENTS

Qualified AIEEE in 2010 to pursue B.Tech.

Qualified GATE in 2015 to pursue M.Tech.

Qualified UGC-NET in 2018 to pursue Ph.D.

SKILLS

Coding

Python C C++

Frameworks and Platforms

Pytorch TensorFlow Scikitlearn
Huggingface Google Colab

RESEARCH INTERESTS

Natural Language Processing
Conversational AI Machine Learning
Deep Learning Reinforcement Learning

STRENGTHS

Hard-working Collaborative
Persuasive Motivator
Ready to learn

LANGUAGES

English
Hindi



NPTEL COURSEWORKS

Natural Language Processing

By Prof. Pushpak Bhattacharyya, IIT Mumbai

Deep Learning

By Prof. Mitesh Khapra, IIT Madras

RESEARCH PAPER WORKS

Published/Accepted

- **Joshi, Ratnesh Kumar**, Arindam Chatterjee, and Asif Ekbal. "Saliency infused dialogue response generation: Improving task-oriented text generation using feature attribution." Expert Systems with Applications (2024): 124283. (h-5 index = 165)
- **Joshi, Ratnesh Kumar**, Arindam Chatterjee, and Asif Ekbal. "Saliency Guided Debiasing: Detecting and mitigating biases in LMs using feature attribution." Neurocomputing 563 (2024): 126851. (h-5 index = 135)
- **Joshi, Ratnesh Kumar**, Arindam Chatterjee, and Asif Ekbal. "Towards explainable dialogue system: Explaining intent classification using saliency techniques." Proceedings of the 18th International Conference on Natural Language Processing (ICON). 2021.
- Priya, Priyanshu, Desai Yasheshbhai, **Ratnesh Joshi**, Roshni Ramnani, Anutosh Maitra, Shubhashis Sengupta, and Asif Ekbal. "TRIP NEGOTIATOR: A Travel Persona-aware Reinforced Dialogue Generation Model for Personalized Integrative Negotiation in Tourism." In Findings of the Association for Computational Linguistics: EMNLP 2024, pp. 16566-16595. 2024.

Under Review

- **Joshi, Ratnesh Kumar**, Saurav Dudhate, Sagnik Sengupta, Mansi Srivastava, Roshni Ramnani, ANUTOSH MAITRA, Asif Ekbal. "Nudging towards Sustainability: Persona-Driven Reinforcement Learning for Empowering Informative Conversation." In ACL (ARR Dec) 2025.
- **Joshi, Ratnesh Kumar**, Priyanshu Priya, Vishesh Desai, Saurav Dudhate, Siddhant Senapati, Asif Ekbal, Roshni Ramnani, Anutosh Maitra, Shubhashis Sengupta. "Strategic Prompting for Conversational Tasks: A Comparative Analysis of Large Language Models Across Diverse Conversational Tasks." Expert Systems with Applications (2024). (h-5 index = 165)

CONFERENCES

20th International Conference on Natural Language Processing (ICON-2021)

[Goa University, India](#)

📅 16 – 19 Dec 2023

📍 Organize - webmaster

18th International Conference on Natural Language Processing (ICON-2021)

[NIT Silchar, India](#)

📅 16 – 19 Dec 2021

📍 Attended - Published

16th International Conference on Natural Language Processing (ICON-2019)

[IIIT Hyderabad, India](#)

📅 18 – 21 Dec 2019

📍 Attended

HOBBIES

Listening Music

Listening Podcasts

Video Games

Watching Movies

MOOC COURSEWORKS

Natural Language Processing Specialization

[By Deeplearning.AI](#)

📅 Aug 2020 – Nov 2020

CS 224

[By Stanford University on Natural Language Processing](#)

📅 July 2019 – Aug 2019

EDUCATION

Ph.D. in Computer Science and Engineering

[Indian Institute of Technology Patna](#)

📅 July 2019 – Ongoing

- **CPI:** 8.3
- **Research Area:** Explainability in Natural Language Generation.
- **Ph.D. Thesis Title:** Investigating Explainability Aided Conversational Systems for Imparting Transparency, Faithfulness, and Trust.

M.Tech.in Computer Science and Engineering

[Central University of Rajasthan](#)

📅 Aug 2015 – June 2017

- **CGPA:** 7.5
- **Research Area:** Text Generation.
- **M.Tech. Thesis Title:** Improving Interpretability of Long Short-Term Memory (LSTM) with Hidden Markov Models (HMM)

B.Tech.in Computer Science and Engineering

[College of Engineering Roorkee](#)

📅 Aug 2010 – June 2014

- **CGPA:** 7.2

REFEREES

Prof. Asif Ekbal

@ Indian Institute of Technology Jodhpur

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<https://ekbalasif.github.io/>

Anutosh Maitra

@ Accenture Labs, Bengaluru

✉ anutosh.maitra@accenture.com